

Sitraka Randrianarivo

Clemson, SC — srandri@clemson.edu

EDUCATIONAL BACKGROUND

- Expected May 2027 **Ph.D. in Mathematical Sciences**
Clemson University, Advisor: Dr. Felice Manganiello
- 2023 **M.S. in Mathematical Sciences**
Clemson University
- 2021 **M.S. in Mathematical Sciences**
African Institute for Mathematical Sciences (AIMS) South Africa
- 2018 **Bachelor's Degree in pure Mathematics**
Faculty of Sciences, University of Antananarivo, Madagascar

PROFESSIONAL APPOINTMENTS

Clemson University, School of Mathematical and Statistical Sciences

- 2025–present **Graduate Teacher of Record**
Courses: Essen Math for Informed Soc (Fall 2025);
Precalc & Intro Differen Calc (Spring 2026);
Engineering Calculus I (Fall 2026).
- 2022–present **Graduate Teaching Assistant**
Global Challenge: Research Experience for Visiting Undergraduate (Summer 2022).
Duties consist on assisting Undergraduate Student on their research.
Course support: Linear Algebra; College Algebra; Pre-calculus
Calculus with one Variable; Calculus of Several Variables;
Business Calculus I; Business Calculus II

PUBLICATIONS

- [1] Felice Manganiello and Sitraka Randrianarivo. Subspace code construction over the complex numbers using extended multiplicative characters. *Submitted*, 2026.
- [2] Felice Manganiello and Sitraka Randrianarivo. Exact cardinality and nonredundant parametrization of character-polynomial codes. *arXiv preprint arXiv:2607.11595*, 2026.
- [3] Sitraka Randrianarivo. Rational Reconstruction for Homological Construction. Master's thesis, African Institute for Mathematical Sciences, South Africa, 2021.

HONORS AND AWARDS

- 2021 **Full Scholarship for M.S. Study**
African Institute for Mathematical Sciences (AIMS)South Africa.

PRESENTATIONS

Research talks and posters

- [1] Felice Manganiello and Sitraka Randrianarivo. Subspace codes from finite fields for the operator channel over complex number. In *2026 Joint Mathematics Meetings (JMM 2026)*. AMS, 2026.
- [2] Sitraka Randrianarivo. Subspace codes for Analogue Operator Channel, 2025. Special-session talk, 2025 Spring Southeastern Sectional Meeting, Clemson University, Clemson, SC.

- [3] Sitraka Randrianarivo. Construction of optimal codes for sum-rank weight, 2024. Poster presentation, Data sECurity and mAchine Learning (DECAL) Workshop, Clemson University, Clemson, SC.

TECHNICAL SKILLS

Mathematical typesetting	$\text{\LaTeX}$
Computer algebra systems	SageMath, Macaulay2, Singular, Mathlab
Programming	Python, C

OUTREACH AND COMMUNITY ENGAGEMENT

2026	Volunteer , Clemson Calculus Challenge (Afternoon Session) Supported the 2026 Clemson Calculus Challenge during its afternoon session.
2024, 2026	Math-In Volunteer , Clemson University Helped students prepare for final exams by answering questions in multivariable calculus (Spring 2024), precalculus, and Calculus I (Spring 2026).

LANGUAGES

English: Fluent **French:** Fluent **Malagasy:** Native